Harrow Creek Water Treatment Plant

Abbreviations

Table A. Abbreviations

Abbreviation	Meaning
CCB	chlorine contact basin
DMF	dual-media filter
FWT	finished water turbidity
HDT	hydraulic detention time
RMB	rapid mix basin
RWP	raw water pump
SB	sedimentation basin
SLR	surface loading rate
SWD	side water depth
TDH	total dynamic head
VG	velocity gradient

1 Design Basis

The Harrow Creek Water Treatment Plant is sized for a design flow of 25.9 MGD (average day) and a peak flow of 42.2 MGD on the maximum day, serving a population of 42,680 persons.

Measured over five years, the 95th percentile raw water turbidity is 37.6 NTU. FWT shall not exceed 0.23 NTU in 95% of monthly samples.

2 Treatment Process

The RMB provides an HDT of 52 s at design flow, with a VG of 922 1/s. Flocculation follows in 4 stages, for a total HDT of 23.5 min at design flow.

There are 2 SBs, each 159 ft long and 47 ft wide, with an SWD of 13.7 ft; the SLR is 0.86 gpm/ft² at design flow.

Filtration uses 8 DMFs of 404 ft² each. With one filter out of service, the filtration rate is 4.1 gpm/ft².

Table 1. RWPs

Tag	Capacity (gpm)	TDH (ft)	Motor (hp)
P-101A	5,151	96.8	267
P-101B	5,590	64.9	139

P-101C	8,542	69.6	232

3 Chemical Feed

Alum is dosed at 38.8 mg/L on the average day and up to 55.7 mg/L on the maximum day. Chlorine is fed at 3.3 mg/L ahead of the clearwell.

4 Disinfection Options

Two options were compared for primary disinfection.

Table 2. Disinfection options

	Option A: UV	Option B: CCB
Capital cost ($M)	6.35	2.82
Annual energy use (MWh/yr)	771	n/a
Contact time at peak flow (min)	n/a	49